



Learning from Mistakes

Older engineering disciplines, such as civil and mechanical engineering, matured over decades of trial and error. Software engineering is still an emerging discipline and its practitioners also need to learn from past failures, as well as from other engineering disciplines to improve its practices.

Compared to civil engineering, which evolved over the last few millennia, the very term 'software engineering' was coined only in 1968, at the first NATO software engineering conference held in Garmisch, Germany. Though electrical and electronics engineering too have matured over time, we cannot wait decades for the software engineering discipline to mature. As software engineers, we can learn from failures, and try to find ways to overcome them.

Many of the failures, accidents and disasters related to software are well-known today. In the past, software failures have had varying impacts, from minor inconveniences and monetary losses to adversely affecting day-to-day life, to even causing the loss of human lives.

Software defects in mobile phones and media players cause minor inconveniences. For example, the Zune media players froze on December 31, 2008, which was the last day of a leap year that the software did not check for. A missing check for validating the input in the onboard software caused the USS Yorktown (a warship in the US Navy) to be stranded at sea for a few hours in 1997. Because of some software defects, Toyota's Prius cars "accelerated unintentionally", causing accidents, following which, thousands of cars were recalled, causing millions in losses to the company. In 1985-87, six people were killed because of an overdose from Therac-25 radiation therapy machines – a result of software malfunctioning. Hence, we can learn from these software failures and also from the history of other engineering disciplines. In fact, some of the earlier disasters have striking similarities with the software disasters that we are familiar with today. Let's look at an example.

The Vasa

Sweden was one of the major military powers in the 1600s. In 1625, Sweden's King Gustav II Adolf ordered a large warship to be built to strengthen the country's naval war capabilities. It was called the Vasa. He assigned this task to an experienced ship-builder named Henrik. Based on the descriptions that the king gave, Henrik designed the ship to carry 32 guns, each weighing 10.88 kg (24 lb). With that plan in mind, he built an approximately 33.52 m (110 feet) keel. Meanwhile, King Gustav learned about Denmark's plan for a larger ship; so he ordered that the Vasa be enlarged so as to have a 41 m (135 feet) keel. Further, he wanted to introduce two enclosed gun decks in the ship, which was an innovative feature for ships of that time.

As the construction of the ship progressed, the king suggested a series of changes, finally deciding that the Vasa must be able to carry 64 guns (each 10.88 kg/24 lb), with many small guns as well. The architect Henrik died before completing the ship, and his assistant Jacobsson was made the architect of the ship. With no written specifications or documents, Jacobsson continued with the ship's construction with whatever limited understanding he had about the original plans. Before launch, a 'stability test' was conducted; the ship rocked violently, and the test was stopped, but fearing the wrath of the king, the architect hid the problems about the ship from the monarch.

When Poland threatened war, the King ordered the ship to be launched as soon as possible. On August 10, 1628, on its maiden voyage, the Vasa sailed only 1,300 metres before it sank, killing 53 sailors. This resulted in a huge economic loss and humiliation to Sweden.

Continued on page no 52